SCIENCE · CLASS 6

Measurement of Length

Notes · why we need standard units, and how to measure correctly

Name: _____

Date: _____

1. Why do we need a standard unit?

Riya and Arjun measured the **same 1.6 m table** with their handspans:



They got different numbers, but **neither is wrong**: Arjun's hand is smaller. Handspans, footspans, cubits and steps change from person to person. So everyone agreed on **standard units** that are the same everywhere.

Remember The **SI unit of length is the metre (m)**. "SI" means the International System of Units, used by the whole world.

2. Bigger and smaller units

Unit	Symbol	Relation	Used for
kilometre	km	1 km = 1000 m	distance between towns
metre	m	1 m = 100 cm	length of a room, cloth
centimetre	cm	1 cm = 10 mm	length of a pencil, book
millimetre	mm	—	thickness of a coin

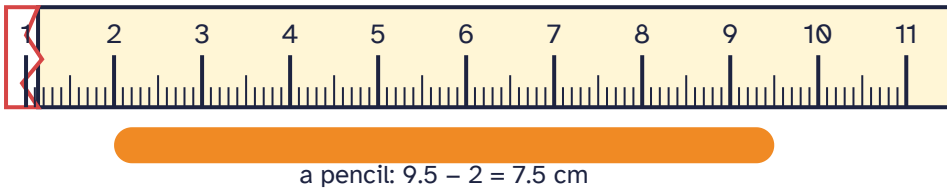
Big unit → small unit: **multiply**. Small unit → big unit: **divide**.

Example 2.5 m = 2.5 × 100 = **250 cm**. 340 cm = 340 ÷ 100 = **3.4 m**.

3. How to measure correctly

- Place the scale **along** the length, touching the object.
- Start from the **zero mark**.
- Keep your eye **straight above** the mark you read. Looking from the side gives a wrong reading.

Broken ruler? Start from any full mark and **subtract**:



a pencil: 9.5 – 2 = 7.5 cm

Common mistake Reading the end mark (9.5 cm) as the length when the object starts at 2 cm.

4. Measuring a curved line

Lay a **thread** carefully along the curve, mark where it ends, then straighten the thread and measure it with a ruler.

Quick summary

- Handspans and steps are not reliable: they differ from person to person.
- SI unit of length: **metre (m)**. 1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm.
- Measure from zero, scale along the object, eye straight above the mark.

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Measurement: worksheet

Name: _____

Try every question. Show your working.

Date: _____

A. Convert the units

1. 2.5 m = ___ cm
2. 340 cm = ___ m
3. 4.2 cm = ___ mm
4. 3 km = ___ m
5. 1500 m = ___ km
6. 75 mm = ___ cm

B. Broken ruler

7. a leaf starts at the **1.5 cm** mark and ends at the **10 cm** mark. How long is it?



C. Handspans

8. The class board is **3.2 m** long. Meena's handspan is 16 cm and Ravi's is 20 cm. How many handspans does each of them count? Who is right?

D. Word problem

9. A running track is **400 m** long. Sana runs **5** laps. How far does she run, in metres and in kilometres?

E. Think

10. Why would a shopkeeper not sell cloth by handspans? Give one reason.

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Measurement: answer key

With short working · every answer checked by code

A. Conversions

1. 2.5 m = **250** cm (big → small: × 100)

2. 340 cm = **3.4** m (small → big: ÷ 100)

3. 4.2 cm = **42** mm (big → small: × 10)

4. 3 km = **3000** m (big → small: × 1000)

5. 1500 m = **1.5** km (small → big: ÷ 1000)

6. 75 mm = **7.5** cm (small → big: ÷ 10)

B. Broken ruler

7. 10 − 1.5 = **8.5 cm**

C. Handspans

8. 3.2 m = 320 cm. Meena: 320 ÷ 16 = **20** spans. Ravi: 320 ÷ 20 = **16** spans. **Both are right**: their hands are different sizes, which is why we need a standard unit.

D. Word problem

9. 5 × 400 m = **2000 m** = 2000 ÷ 1000 = **2 km**

E. Think

10. Sample answer: different people have different handspans, so customers would get different amounts of cloth for the same price. A metre is the same for everyone.

Original questions by Adithya Vikram · matches the Class 6 Short "Who measured it wrong?" · every numerical answer computed and double-checked by code.